IMPORTANT FUNCTIONS & PROBABILITY DISTRIBUTIONS

Probability Distribution	p.m.f or p.d.f	Mean	Variance
Uniform (discrete)	$P(X=x)=1/N, \quad N=1,2,\ldots$	$E(X)=(N+1)/2$	$V(X)=(N^2-1)/12$
Bernoulli	$P(X=x)=p^x q^{1-x}, \quad x=0,1$	$E(X)=p$	$V(X)=pq$
Binomial	$P(X=x)={}^nC_x p^x q^{n-x}, \quad x=0,1,2,\ldots,n$	$E(X)=np$	$V(X)=npq$
Poisson	$P(X=x)=\frac{e^{-\lambda}.\lambda^x}{x!}, \quad x=0,1,2,\ldots,\infty$	$E(X)=\lambda$	$V(X)=\lambda$
Geometric	$P(X=x)=pq^{x-1}, \quad x=1,2,3,\ldots,\infty$	$E(X)=1/p$	$V(X)=q/p^2$
Negative Binomial	$P(X=x)={}^{x-1}C_{r-1}p^r q^{x-r}, \quad x=r,r+1,r+2,\ldots$	$E(X)=r/p$	$V(X)=rq/p^2$
Uniform (continuous)	$f(x)=1/b-a, \quad a\le x\le b$	$E(X)=(a+b)/2$	$V(X)=(a-b)^2/12$
Exponential	$f(x)=\lambda e^{-\lambda x}, \quad x\ge 0$	$E(X)=1/\lambda$	$V(X)=1/\lambda^2$
Gamma	$f(x)=\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}, \quad x>0$	$E(X)=\alpha/\lambda$	$V(X)=\alpha/\lambda^2$
Beta	$f(x)=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha).\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}, \quad 0<x<1$	$E(X)=\alpha/\alpha+\beta$	$V(X)=\frac{\alpha.\beta}{(\alpha+\beta)^2.(\alpha+\beta+1)}$
Normal	$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty<x<\infty$	$E(X)=\mu$	$V(X)=\sigma^2$

GAMMA FUNCTION

$$\Gamma(n)=\int_0^\infty x^{n-1}e^{-x}dx \qquad \text{Also} \qquad \int_0^\infty x^{n-1}e^{-ax}dx=\frac{\Gamma(n)}{a^n}$$

BETA FUNCTION

$$B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx=\frac{\Gamma(\alpha).\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

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